

ARNAV SINGH

Machine Learning Engineer | LLM Fine-Tuning & Model Evaluation

+91 7428775691 | arnavwork08@gmail.com | [LinkedIn](#) | [GitHub](#) | [Kaggle](#) | [Portfolio](#)

SUMMARY

Machine Learning Engineer and final-year B.Tech Computer Engineering student (CGPA 8.45) who builds production ML systems end to end: data pipelines, model training, experiment tracking, model validation and cloud deployment as REST APIs. Fine-tuned Mistral-7B with QLoRA (~75% lower training memory) and BERT for legal NLP, and shipped a fraud detection model (AUC-PR 0.9752) as a live FastAPI inference service. Differentiator: rigorous model evaluation, with a focus on LLM benchmarking.

TECHNICAL SKILLS

Machine Learning & Data: Python, SQL, Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, Feature Engineering, EDA, Imbalanced Learning (SMOTE, ADASYN), statsmodels, Power BI

LLM & NLP: Hugging Face Transformers, PEFT (LoRA, QLoRA), BitsAndBytes 4-bit Quantization, Mistral-7B, BERT Fine-Tuning, FAISS, BERTScore, Prompt Engineering

Model Evaluation: Model Validation, Ablation Studies, Hyperparameter Tuning, AUC-PR, Precision/Recall/F1, Perplexity, Inference Latency Benchmarking, SHAP, Conformal Prediction

MLOps & Deployment: FastAPI, Docker, MLflow (Experiment Tracking), Weights & Biases, Render, AWS SageMaker (basics), Streamlit, Git

EXPERIENCE

Data Analyst Intern — Jambudweep Shodh Kendra Pvt Ltd

Aug 2026 – Present

- Analyze multi-year electoral datasets (constituency, Lok Sabha, Rajya Sabha) for voting-pattern shifts; validate predicted trends against historical results with statistical correlation tests.

PROJECTS

Credit Card Fraud Detection — ML Pipeline & Live API | XGBoost, SMOTE, SHAP, MLflow, FastAPI, Docker | [Live Demo](#) · [GitHub](#)

- Built an end-to-end ML pipeline on 284K transactions (<1% fraud rate): data preprocessing, SMOTE resampling and XGBoost training, with experiment tracking in MLflow.
- Achieved AUC-PR 0.9752, 87.5% recall and a 0.15% false-alarm rate; estimated £5,338 saved versus a no-detection baseline.
- Added SHAP global and transaction-level explanations; containerized with Docker and deployed on Render as a REST API with a real-time inference endpoint.

LegalSimplify — Legal Clause Simplifier (Mistral-7B, QLoRA) | Hugging Face, PEFT, LoRA, QLoRA, BitsAndBytes | [GitHub](#)

- Fine-tuned Mistral-7B-Instruct on a custom legal dataset using LoRA with 4-bit quantization, cutting training memory by ~75%.
- Ran an ablation study over LoRA ranks 4, 8, 16, 32 and 64, benchmarking perplexity, inference latency and task quality; selected rank 16 as the best trade-off for the legal domain.

Legal Query Classifier — BERT Inference Service | BERT, PyTorch, FastAPI, Docker | [GitHub](#)

- Fine-tuned BERT for multi-task legal text classification, reaching 90%+ accuracy; handled class imbalance with weighted loss.
- Evaluated with per-class precision, recall and F1, and documented failure modes for model monitoring and future iteration.
- Containerized with Docker and served through FastAPI for production-style model inference.

AI-Based Fake Identity & Document Screening System | OpenCV, EasyOCR/Tesseract, DeepFace/InsightFace, Streamlit | [GitHub](#)

- Designed a five-stage computer vision pipeline (OCR → validation → tamper detection → face verification → risk scoring) using pretrained models, ICAO 9303 MRZ check-digit parsing and Error Level Analysis.
- Combined face-embedding similarity with a tunable weighted risk score to return real-time PASS/FLAG verdicts in a Streamlit dashboard.

EDUCATION

B.Tech, Computer Engineering — ABES Engineering College (AKTU)

2023 – 2027

CGPA: 8.45 | Coursework: Machine Learning, Probability & Statistics, Data Structures & Algorithms, DBMS, Operating Systems

ACHIEVEMENTS & CERTIFICATIONS

- 3rd Place, Retina AI Hackathon 2026; 2nd Place, ECell Eureka Competition Zonals (IIT Bombay ECell).
- Solved 300+ DSA problems on [LeetCode](#) and [CodeChef](#); CodeChef rating 1614 (3 Star).
- [2025 Google Solution Challenge Participant](#); [Google 5-Day Agentic AI Vibe Coding Contributor](#).
- Certifications: Data Science & ML Fundamentals; Regression & Prediction with Pandas and Python; SQL for Data Analytics; IBM SkillsBuild Data Fundamentals.

LEADERSHIP

- Growth & Content Strategist, Aasira** (Jul 2024 – Feb 2026): +35% engagement; **Volunteer, GDGoC** (2024 – 25): +40% event attendance.